



Lecture 3: Planetary Heat & Energy Transport

Planetary Systems

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Learning objectives

By the end of this lecture, you will be able to:

- Identify the main energy sources driving planetary evolution
- Distinguish conduction, convection, and radiation as heat transport mechanisms
- Derive and apply the conductive cooling timescale $\tau \sim L^2/\kappa$
- Explain the Rayleigh and Nusselt numbers and what they tell us
- Describe tidal heating and why Io is the most volcanic body in the solar system

Recap: Lecture 2

- Protoplanetary disks: structure, snow line, 3–5 Myr lifetime
- Dust → planetesimals → planets (streaming instability, pebble accretion)
- Core accretion vs. gravitational instability for giant planets
- Kepler's laws, vis-viva equation
- Orbital resonances, tidal forces, migration

Today: What happens *inside* planets after they form?

A grayscale image of the planet Io, showing its heavily cratered surface. A bright, elongated plume of material is visible on the left side of the planet, extending into the dark space. The text "1. Energy sources" is overlaid in white on the right side of the planet.

1. Energy sources

What heats a planet?

Four principal sources of internal heat:

1. **Accretional heating:** gravitational energy from assembly
2. **Gravitational differentiation:** core formation releases ~10% more
3. **Radioactive decay:** sustained for billions of years
4. **Tidal heating:** orbital energy dissipated as friction

Without internal heat → no volcanism, no magnetic field, no plate tectonics, no atmospheric outgassing.

Accretional heating

Gravitational energy of assembling a uniform sphere:

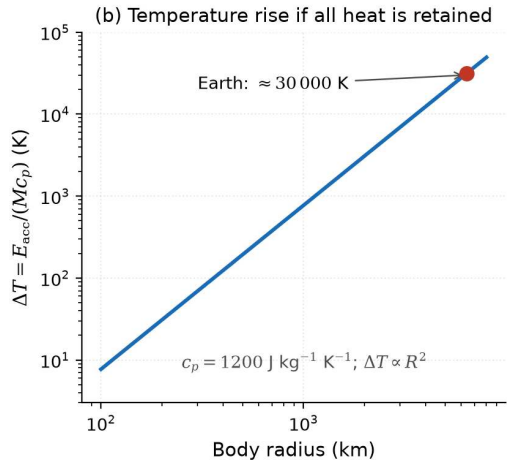
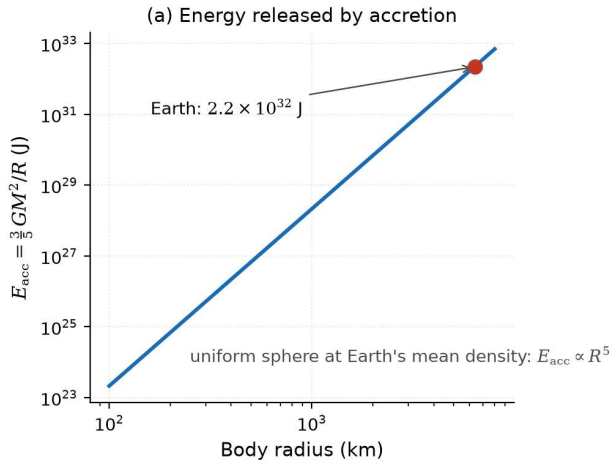
$$E_{\text{acc}} \sim \frac{3}{5} \frac{GM^2}{R}$$

For Earth: $E_{\text{acc}} \approx 2.2 \times 10^{32} \text{ J}$

If all retained as heat:

$$\Delta T = \frac{E_{\text{acc}}}{Mc_p} \approx \frac{2.2 \times 10^{32}}{5.97 \times 10^{24} \times 1200} \approx 30,000 \text{ K}$$

Enough to melt Earth many times over. Accretion inevitably produces a **magma ocean**.



Accretional heating against body size for a uniform sphere at Earth's mean density: $E_{\text{acc}} \propto R^5$ and the retained temperature rise $\propto R^2$, about 30 000 K for Earth but tens of kelvins for radii of 100 to 300 km. Course figure.

Where does the accretional energy go?

Not all E_{acc} is retained as interior heat.

- **Radiation to space:** hot surface cools efficiently during late accretion
- **Thermal blanketing:** a thick primordial atmosphere traps heat, raising retention
- **Depth of deposition:** energy from giant impacts is buried deep, while small impacts deposit near the surface
- Typical retained fraction: 10–50% (depends on impact size distribution)

Even 10% retention is enough to fully melt a terrestrial planet: the magma ocean is effectively guaranteed for bodies $\geq$ Mars-sized.

Where do heat-producing elements live?

Preview for Lecture 4: elements partition differently between metal and silicate phases.

- **Lithophile** (“rock-loving”): partition into silicate phase
 - U, Th, K: the long-lived heat producers ⇒ **mantle and crust**
- **Siderophile** (“iron-loving”): partition into metallic iron
 - Fe, Ni, Co, Pt, Ir, Au ⇒ **core**; very little radiogenic heating there today

Because U, Th, K are lithophile, radiogenic heat is produced in the **mantle**, not the core, and that sustains mantle convection.

Source: Goldschmidt (1937); McDonough (2003)

Gravitational differentiation

A second, one-time burst of heat accompanies **core formation**.

- Dense iron droplets sink from the magma ocean to the centre
- Gravitational potential energy → heat via viscous friction
- Energy release ~10% of the accretional budget

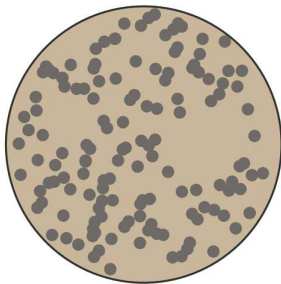
Order-of-magnitude estimate for Earth:

$$E_{\text{diff}} \sim \frac{3}{10} \frac{GM_{\text{core}}^2}{R_{\text{core}}} \left(\frac{R}{R_{\text{core}}} - 1 \right) \approx 2 \times 10^{31} \text{ J}$$

This heat is released **once**, during core formation, and sustains the magma ocean long enough for chemical differentiation (Lecture 4).

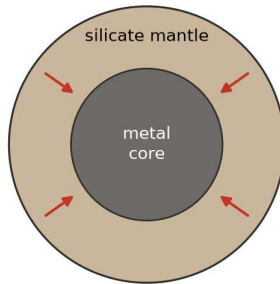
Gravitational differentiation (schematic)

(a) Homogeneous body after accretion



metal droplets dispersed in silicate rock

(b) After core formation



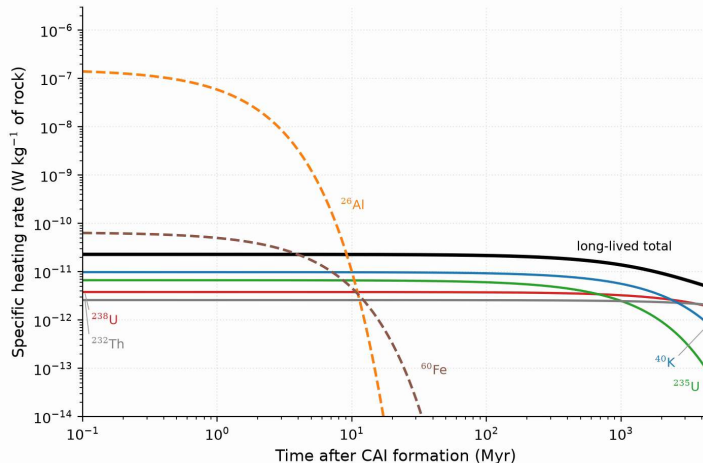
dense metal sinks: $E_{\text{diff}} \approx 2 \times 10^{31}$ J for Earth

Gravitational differentiation: metal droplets dispersed in rock after accretion (a) sink to form a core (b); the drop in potential energy releases about 2×10^{31} J for Earth. Course schematic.

Radioactive decay: the long-lived engine

Isotope	Half-life (Gyr)	Heat (W kg^{-1})	Concentration
^{238}U	4.47	9.5×10^{-5}	20 ppb
^{235}U	0.704	5.7×10^{-4}	0.14 ppb
^{232}Th	14.0	2.6×10^{-5}	80 ppb
^{40}K	1.25	2.9×10^{-5}	28 ppb

- Lithophile elements → concentrate in mantle/crust
- Present-day radiogenic heat: ~20 TW
- 4.5 Gyr ago: almost 5× higher (factor 4.7 with these abundances)

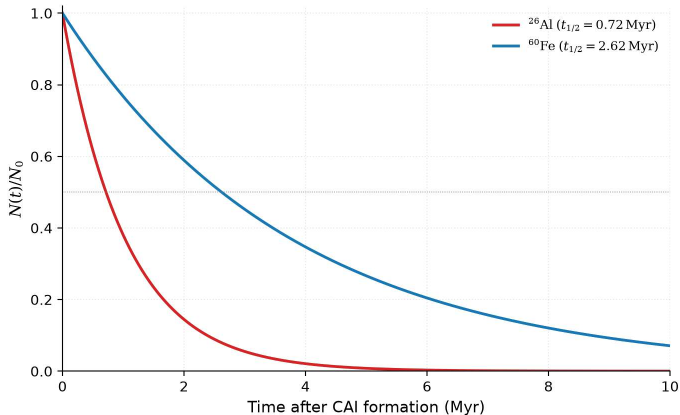


Two heating clocks: ²⁶Al (dashed) starts near 10⁻⁷ W kg⁻¹ and dies within about 10 Myr; ⁶⁰Fe is three orders of magnitude weaker; the four long-lived isotopes (solid) give 92 TW at formation and 19 TW today when integrated over the bulk silicate Earth, a factor 4.7. The crossover near 10 Myr hands the heating from planetesimals to planets. Course figure.

Short-lived isotopes: ^{26}Al

- Half-life: only 0.72 Myr (now extinct)
- Produced by nearby stellar nucleosynthesis, delivered into the protoplanetary disk
- Initial $^{26}\text{Al}/^{27}\text{Al}$ ratio $\sim 5 \times 10^{-5}$ (canonical)
- In the first few Myr: extraordinarily potent heat source ($10^{-7} \text{ W kg}^{-1}$ initially)
- Could melt planetesimals as small as a few km radius
- Initiated core formation and volatile loss **before** planets fully assembled

^{26}Al is why even small asteroids show evidence of melting and differentiation in the meteorite record.



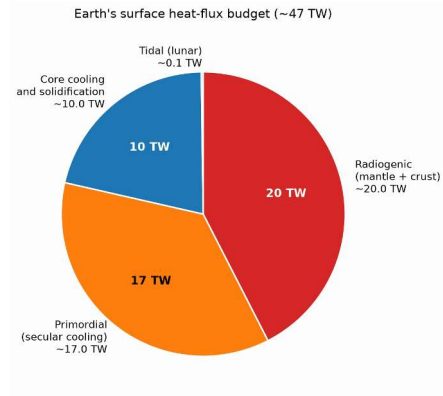
Decay of ^{26}Al (half-life 0.72 Myr, daughter ^{26}Mg) and ^{60}Fe (2.6 Myr, ^{60}Ni) over the first 10 Myr. The other extinct clocks, ^{53}Mn – ^{53}Cr (3.7 Myr), ^{182}Hf – ^{182}W (8.9 Myr), ^{129}I – ^{129}Xe (16 Myr) and ^{244}Pu – ^{244}Xe (81 Myr), date early differentiation, core formation, atmosphere retention and the closure of mantle degassing. ^{26}Al came from a nearby massive star or supernova, so the Sun formed in a massive star cluster. Course figure.

Earth's heat budget: 47 TW

Source	Contribution
Radiogenic heating (mantle + crust)	~20 TW
Primordial (secular cooling)	~17 TW
Core cooling and solidification	~10 TW
Tidal dissipation (from the Moon)	~0.1 TW
Total	~47 TW

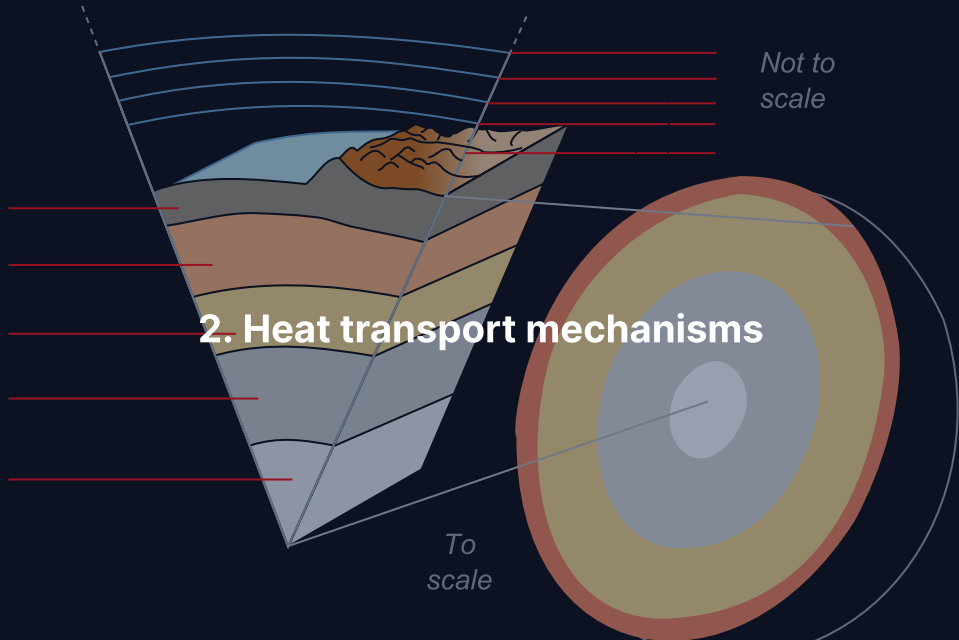
- ~40% from ongoing radioactive decay; 60% from primordial heat.
- Earth is still cooling.

Earth's heat budget at a glance



- Radiogenic decay (red, 20 TW) and stored primordial heat (orange, 17 TW) carry most of the 47 TW.
- Core cooling and inner-core solidification add 10 TW; lunar tidal dissipation, at 0.1 TW, is invisible at this scale.
- The balance shifts over time: 4.5 Gyr ago the radiogenic slice alone was almost 5 times larger.

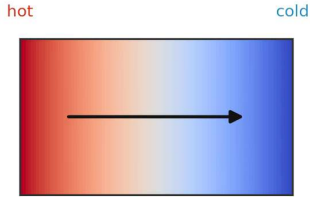
Earth today runs roughly 40% on radioactivity and 60% on its savings account of primordial heat.



Three ways to move heat

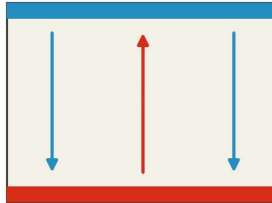
- Conduction:** Heat transferred by molecular and atomic collisions, $\vec{q} = -k\nabla T$ (Fourier's law); slow and diffusive; dominant in lithospheres and small bodies.
- Convection:** Heat transported by bulk fluid motion: hot material rises, cool material sinks; dominant in mantles, outer cores and giant-planet interiors.
- Radiation:** Energy carried by electromagnetic waves, $F = \epsilon\sigma T^4$ (Stefan-Boltzmann); dominant at surfaces, in thin atmospheres and in magma oceans above about 2000 K; solid silicates are opaque in the infrared.

(a) Conduction



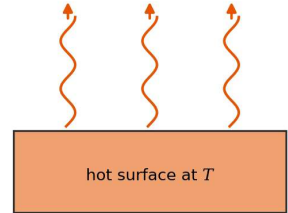
$$\vec{q} = -k\nabla T: \text{no bulk motion}$$

(b) Convection



hot fluid rises, cold fluid sinks

(c) Radiation



$$F = \epsilon\sigma T^4: \text{photons carry the energy}$$

Three ways to move heat: conduction through a material down a temperature gradient without bulk motion, convection by rising hot and sinking cold fluid, radiation by photons from a hot surface. Course schematic.

Conduction: Fourier's law and the diffusion timescale

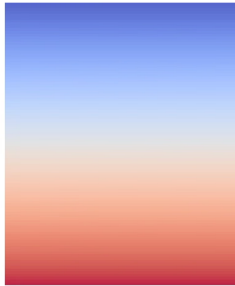
- Heat flux $\vec{q} = -k\nabla T$
- Thermal diffusivity
 $\kappa = k/(\rho c_p) \approx 10^{-6} \text{ m}^2 \text{ s}^{-1}$ for rock
- Metals have high k but also high ρc_p ; rocks have low k and low ρc_p : similar κ

$$\tau_{\text{cond}} \sim \frac{L^2}{\kappa}$$

Material	k (W m ⁻¹ K ⁻¹)	κ (m ² s ⁻¹)
Silicate rock	3–5	$\sim 10^{-6}$
Bridgmanite	5–10	$\sim 2 \times 10^{-6}$
Water ice	2.2	1.1×10^{-6}
Iron (solid)	80	$\sim 2 \times 10^{-5}$
Air	0.025	$\sim 2 \times 10^{-5}$

(a) Fourier's law in a slab

cold surface T_s

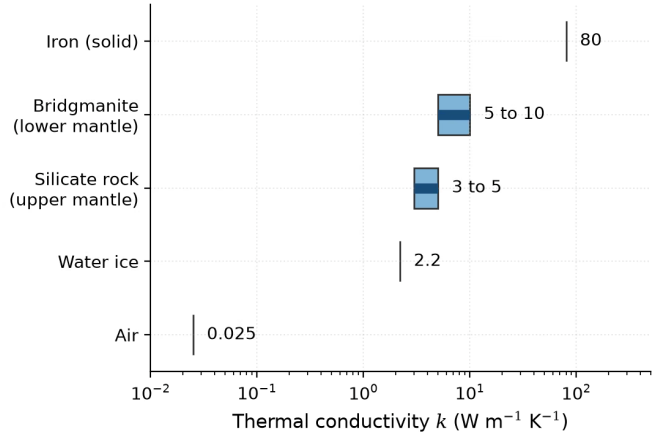


$$q = -k \frac{dT}{dz}$$

heat flows down
the gradient

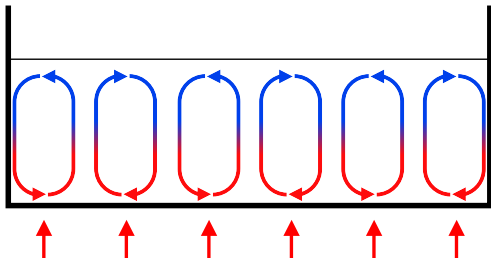
hot interior T_i

(b) Conductivity of planetary materials



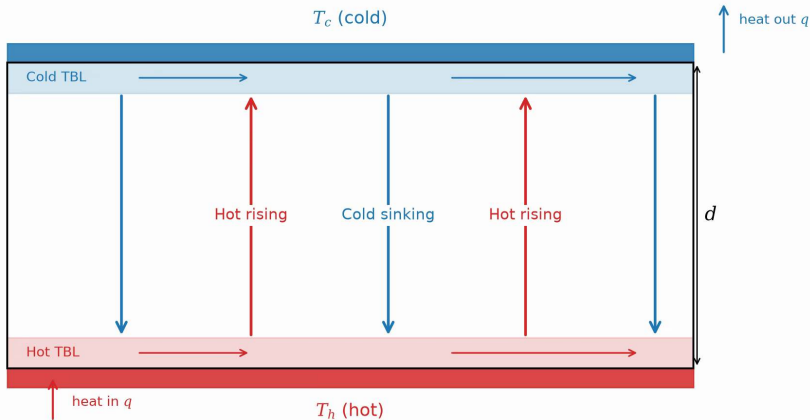
Fourier's law in a slab, and the thermal conductivity of planetary materials from air (0.025) to solid iron ($80 \text{ W m}^{-1} \text{K}^{-1}$). Course figure.

Convection: nature's heat elevator

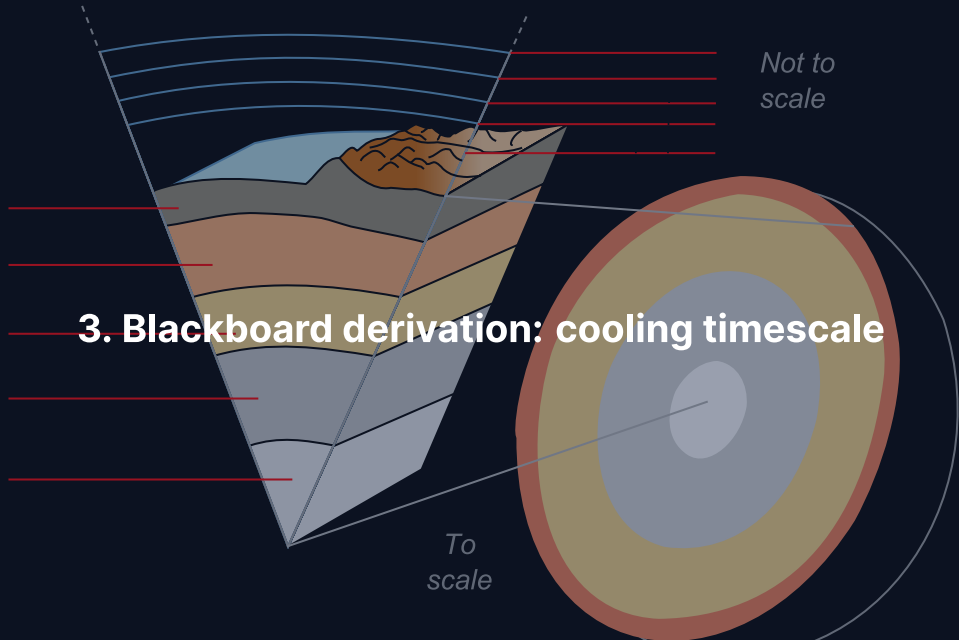


Credit: Wikimedia Commons, CC BY-SA 3.0

- Hot material at base: buoyant, rises
- Cool material at top: dense, sinks
- Far more efficient than conduction
- Active in:
 - Earth's mantle (~cm/yr)
 - Outer core (liquid metal)
 - Giant planet interiors
 - Atmospheres



Rayleigh-Bénard convection: hot plumes rise from a thin basal boundary layer, cold fluid sinks from the top one, and lateral flow closes the cells. Convection switches on once Ra exceeds Ra_c . Thin conductive skins around a well-mixed interior: that is mantle convection in one sketch. Course schematic.



Goal and strategy

Goal: Estimate how long it takes heat to conduct out of a body of size L .

Strategy: Start from the heat equation, then use dimensional analysis instead of solving the PDE.

- We want a characteristic timescale $\tau_{\text{cond}}(L, \kappa)$
- Plug in representative values for rock
- Compare to the age of the solar system (4.6 Gyr)

Physical intuition: bigger bodies have a longer path for heat to diffuse, so they stay hot for longer. We will derive exactly *how much* longer.

Step 1: the heat diffusion equation

Fourier's law $\vec{q} = -k\nabla T$ combined with energy conservation gives:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T)$$

For constant thermal conductivity k , define $\kappa = k/(\rho c_p)$:

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T$$

In one dimension:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

The rate of temperature change is proportional to the **curvature** of the temperature profile; flat profiles do not evolve, curved ones relax.

Step 2: dimensional analysis

Replace derivatives with characteristic scales:

- Time derivative: $\frac{\partial T}{\partial t} \sim \frac{T}{\tau}$
- Spatial derivative: $\frac{\partial^2 T}{\partial x^2} \sim \frac{T}{L^2}$

Substitute into the heat equation:

$$\frac{T}{\tau} \sim \kappa \frac{T}{L^2}$$

Temperature cancels: the timescale is intrinsic, not tied to any particular T .

$$\tau_{\text{cond}} \sim \frac{L^2}{\kappa}$$

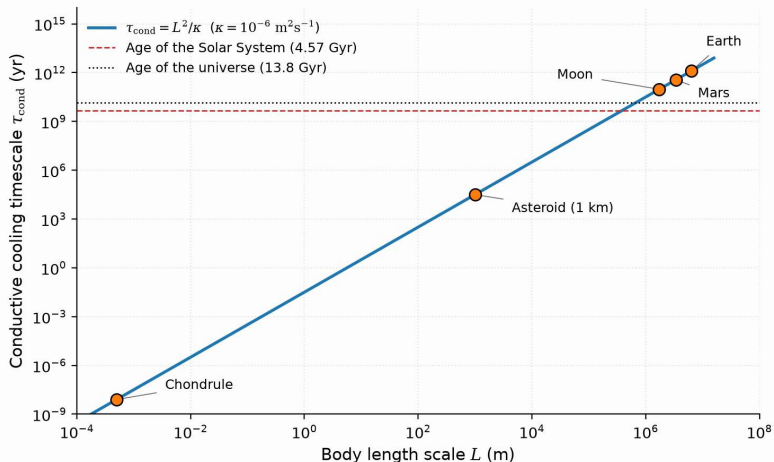
Key feature: the L^2 scaling. Doubling the body size $\rightarrow$ 4 $\times$ the cooling time.

Cooling timescales across the solar system

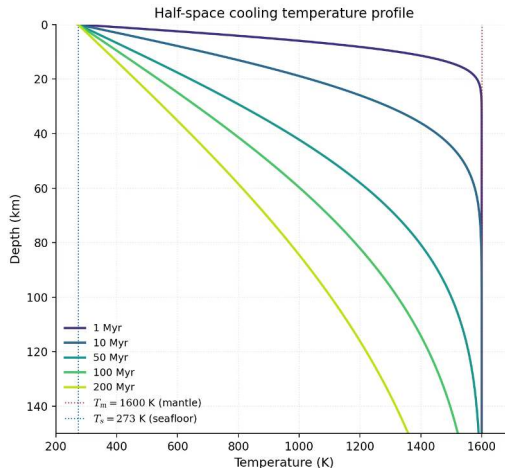
Using $\kappa \approx 10^{-6} \text{ m}^2 \text{ s}^{-1}$ (silicate rock):

Body	Size L	$\tau = L^2 / \kappa$	Familiar units
1 m boulder	1 m	10^6 s	~12 days
100 km asteroid	10^5 m	10^{16} s	~300 Myr
Moon	$1.7 \times 10^6 \text{ m}$	$3 \times 10^{18} \text{ s}$	~100 Gyr
Earth	$6.4 \times 10^6 \text{ m}$	$4 \times 10^{19} \text{ s}$	~1300 Gyr

Earth's conductive cooling time $\gg$ age of the universe.
Yet Earth loses 47 TW $\Rightarrow$ there **must** be convection.

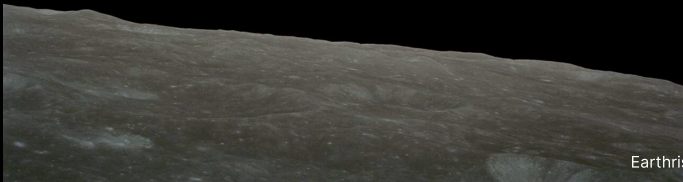


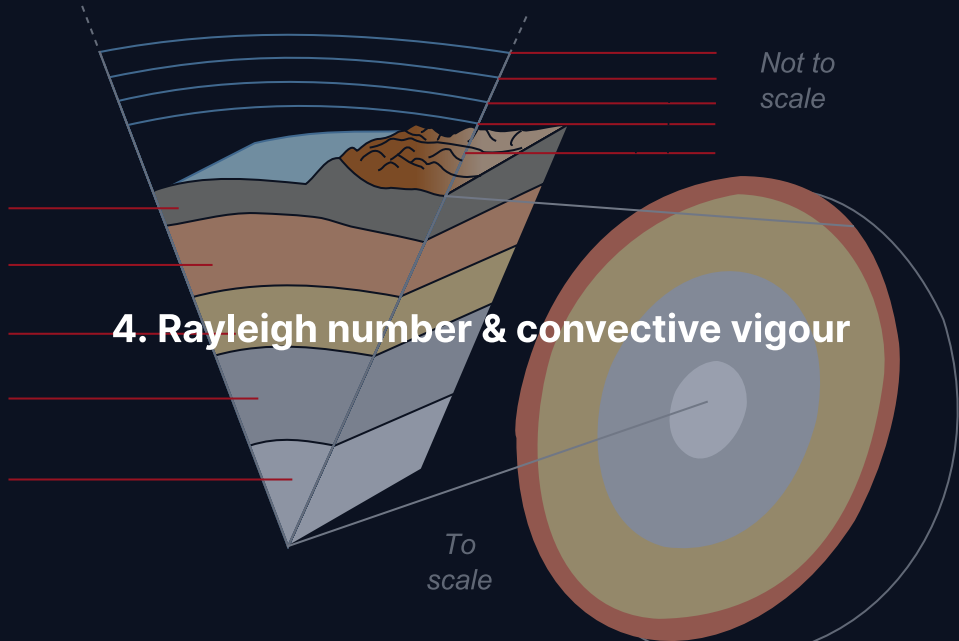
The conductive cooling time $\tau = L^2/\kappa$ against body size (slope 2 on log axes): a 0.5 mm chondrule cools in under a second, a 1 km asteroid in about 30 000 yr; the Moon, Mars and Earth would need longer than the age of the universe. Any body larger than a few hundred km must convect, or stay hot. Course figure.



Half-space cooling: each curve is the error-function solution $T(z, t) = T_s + (T_m - T_s) \text{erf}(z/2\sqrt{kt})$ for ages 1 to 200 Myr. The lithosphere thickens as $\sqrt{kt}$, the heat flux falls as $t^{-1/2}$, and the seafloor deepens with age: one solution explains lithosphere thickness, seafloor heat flow and bathymetry. Course figure, after Turcotte & Schubert (2002).

Break





The Rayleigh number

Does a fluid layer convect or merely conduct?

$$Ra = \frac{\alpha \rho g \Delta T d^3}{\kappa \eta}$$

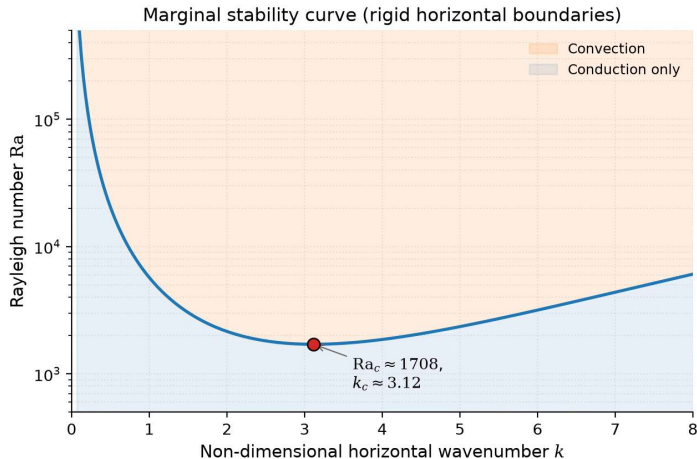
Numerator (buoyancy):

- α = thermal expansion
- ρ = density
- g = gravity
- ΔT = temperature contrast
- d = layer depth

Denominator (dissipation):

- κ = thermal diffusivity
- η = dynamic viscosity

Critical value: $Ra_c \approx 1708$
(rigid boundaries: fluid sticks to wall;
658 for free-slip: fluid slides without
friction)



Marginal stability: below the curve every perturbation decays and conduction is the only steady state; above it convection grows. The minimum is $Ra_c \approx 1708$ at wavenumber $k_c \approx 3.12$: convection is a threshold with a preferred cell size set by the layer depth. Course figure, after Turcotte & Schubert (2002).

Earth's mantle: vigorous convection, and how efficient

$$Ra_{\oplus} \sim \frac{2 \times 10^{-5} \times 4000 \times 10 \times 2500 \times (3 \times 10^6)^3}{10^{-6} \times 10^{21}} \sim 10^7 - 10^8 \gg Ra_c \approx 1708$$

- The Nusselt number $Nu = q_{\text{total}}/q_{\text{cond}}$ measures the efficiency: $Nu = 1$ is pure conduction.
- Boundary-layer theory gives $Nu \propto Ra^{1/3}$: doubling Ra adds only about 26%; a factor 10^3 in Ra gives a factor 10 in Nu .

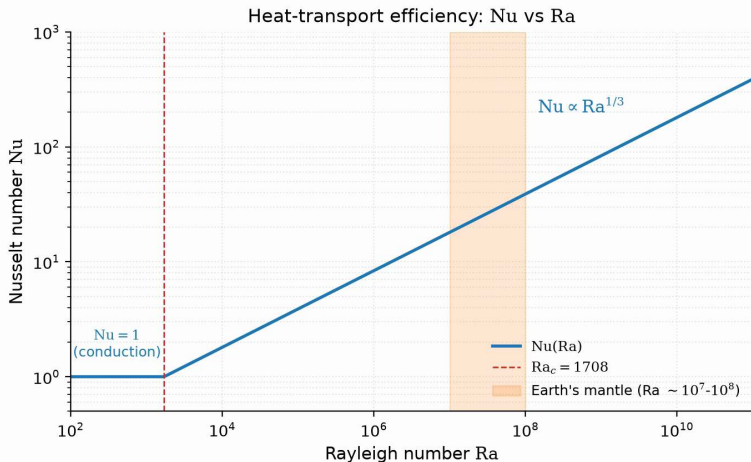
For Earth, $Nu = (Ra/Ra_c)^{1/3} \approx 18-39$: convection moves heat tens of times faster than conduction, and the interior cools efficiently.

Source: Schubert et al. (2001), *Mantle Convection in the Earth and Planets*

Rayleigh number across the solar system

Body	Ra	Regime
Earth's mantle	10^7 – 10^8	Vigorous convection
Venus' mantle	10^7 – 10^8	Vigorous (stagnant lid)
Mars' mantle	10^6 – 10^7	Moderate convection
Moon's mantle	10^4 – 10^5	Weak; cooling by conduction
Mercury's mantle	$\sim 10^4$	Marginal; largely cooled
Magma ocean	$\geq 10^{25}$	Turbulent convection
Earth's outer core	$\geq 10^{25}$	Turbulent (dynamo)

Ra is hyper-sensitive to size ($\propto d^3$) and viscosity ($\propto 1/\eta$). Small bodies and very viscous interiors quickly drop below Ra_c .



$Nu = 1$ below Ra_c ; above it $Nu \propto Ra^{1/3}$. Earth's mantle at $Ra \sim 10^7$ to 10^8 (shaded) moves heat 18 to 39 times faster than conduction, but the weak $1/3$ power means the efficiency grows only gently with the driving: even vigorous mantles need Gyr to cool. Course figure.

Thermal boundary layers

A convecting interior develops three layers:

1. **Cold boundary layer (top):** conductive, steep T drop
→ Earth's **lithosphere** (~100–200 km)
2. **Well-mixed interior:** nearly adiabatic, slow T variation
→ bulk of the mantle
3. **Hot boundary layer (bottom):** steep T rise into the core
→ Earth's **D" layer** (pronounced "D double prime"; ~200 km)

Nearly all the temperature drop occurs in the two thin boundary layers.

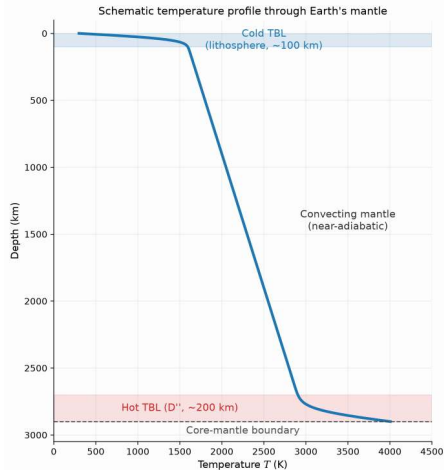
The adiabatic gradient

In the well-mixed convecting interior, temperature follows an **adiabat**:

$$\left(\frac{dT}{dr}\right)_{\text{adiabat}} = -\frac{\alpha g T}{c_p}$$

- α = thermal expansion, g = gravity, c_p = specific heat
- For Earth's upper mantle: ~0.3–0.5 K/km (a fraction of a K per km)
- Much shallower than the conductive gradient in the lithosphere (~25 K/km)
- Adiabatic profiles are nearly isothermal on the scale of the mantle

Temperature changes only a few hundred K over the entire convecting mantle: efficient mixing keeps the interior nearly adiabatic.



Earth's mantle temperature: a cold boundary layer (the lithosphere, about 100 km) with a steep conductive gradient, a near-adiabatic interior at about 0.5 K km^{-1} , and the hot D'' layer (about 200 km) that bridges the 2500 K adiabat to the 4000 K core-mantle boundary. Almost the whole temperature drop sits in two thin skins. Schematic after Schubert et al. (2001).

Two tectonic regimes

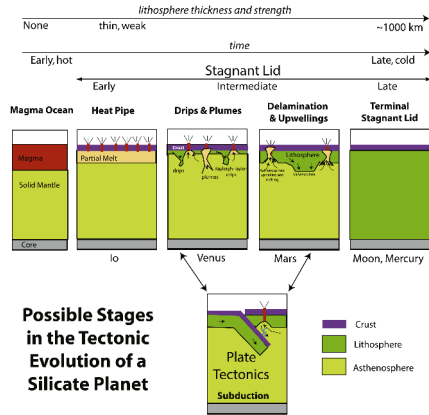
Mobile lid (plate tectonics):

- Cold boundary layer **breaks** into plates
- Plates subduct, recycle material
- Most efficient heat loss
- **Only Earth**

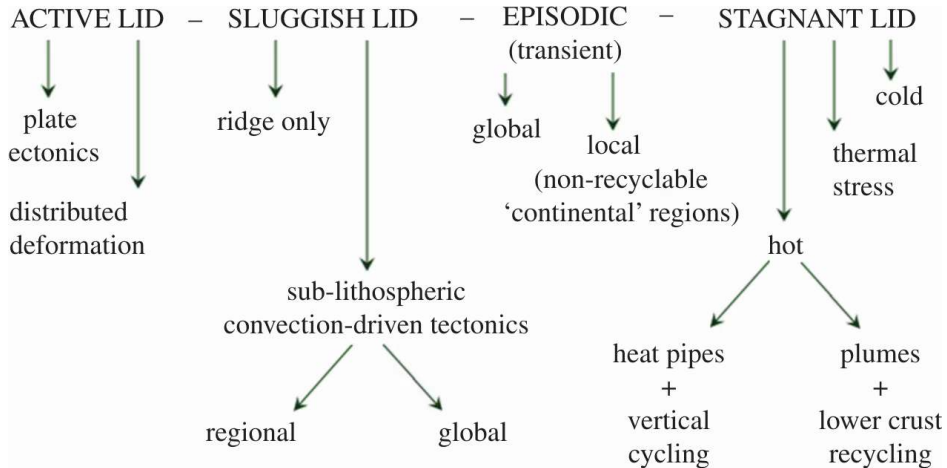
Stagnant lid:

- Single thick, immobile shell
- Heat escapes by conduction + volcanism
- Less efficient cooling
- Venus, Mars, Mercury, Moon

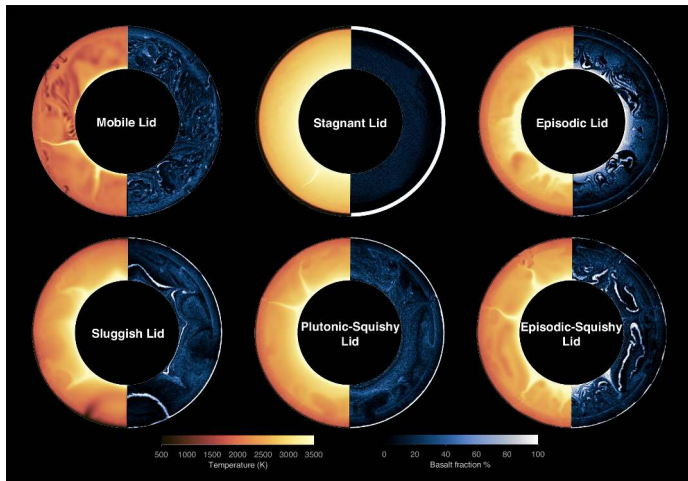
Why only Earth: likely water, size, and thermal history together.



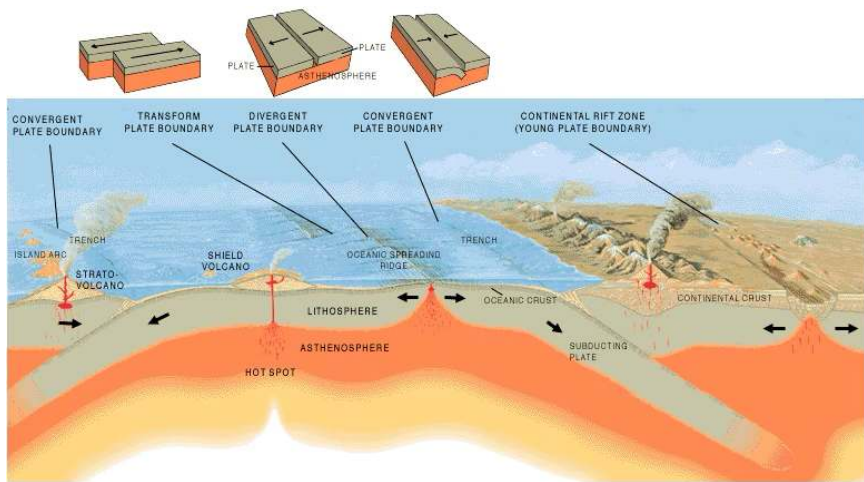
Credit: Reproduced from Stern et al. (2018), Fig. 3



The taxonomy of tectonic modes: active lids (plate tectonics, ridge-only, distributed deformation), episodic modes, and stagnant lids in cold and hot (heat-pipe, plume-fed) variants. Mobile and stagnant are the two end-members of a spectrum that a planet can move through as it cools. Reproduced from Lenardic (2018), Fig. 2.



Six tectonic lid regimes in numerical mantle convection simulations (temperature left, basalt fraction right in each annulus): mobile, stagnant, episodic, sluggish, plutonic-squishy and episodic-squishy lid. Reproduced from Lyu et al. (2025), Fig. 1, CC BY 4.0.

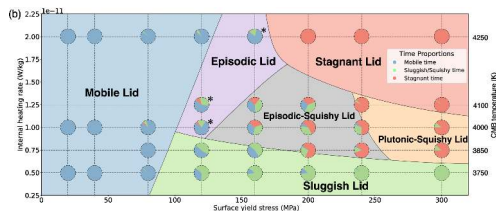


The plate-tectonic settings in one cross-section: ridges where mantle upwells and new crust forms, convergent margins where the cold plate subducts and feeds arc volcanism, and a plume rising independently of the plates. The lithosphere is the cold thermal boundary layer of mantle convection, and subduction is its cold downwelling. Credit: USGS, public domain.

Why does Earth have plate tectonics?

One of the biggest open questions in planetary geophysics. Proposed factors:

- **Water:** weakens faults; lowers the melting point in subducting slabs
- **Planet size:** higher convective stresses in a larger mantle
- **Thermal state:** cold enough for a rigid lid, hot enough for rifting
- **Prior history:** once started, recycling weakens the lid; may be self-sustaining
- Venus: deformed tessera terrain records past plate tectonics, lost after drying out



Credit: Simulated tectonic regimes vs. lithosphere strength and internal heating; reproduced from Lyu et al. (2025), Fig. 5b, CC

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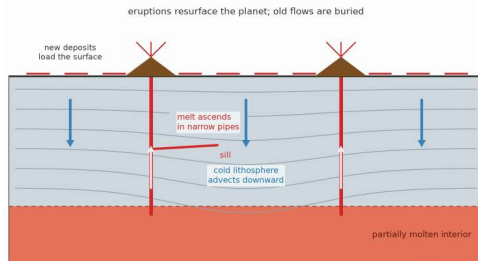
Habitability-oriented view: plate tectonics is linked to liquid water, and both are linked to long-term climate stability.

Heat-pipe volcanism

An alternative heat-loss regime for very hot bodies.

- Magma rises through conduits (“heat pipes”)
- Erupts on the surface, cools, and is buried by new lava
- Old crust subsides into the hot interior as new lava covers it
- **lo today:** global resurfacing by volcanism every $\sim 10^6$ yr
- **Early Earth:** may have been a heat-pipe regime (Hadean, >4 Ga)
- Allows efficient cooling of a very hot mantle without plate tectonics

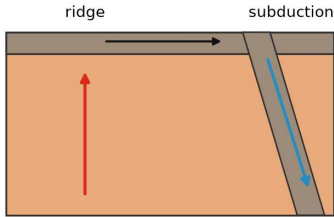
Source: Turcotte (1989); Moore & Webb (2013), *Nature*



Credit: Custom schematic, after Moore & Webb (2013)

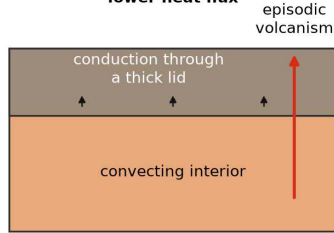
(a) Mobile lid (plate tectonics)

high heat flux



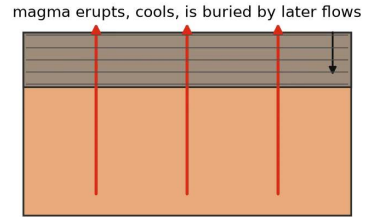
(b) Stagnant lid

lower heat flux



(c) Heat pipe (Io, early Earth)

high heat flux without plates



Heat loss under three regimes: a mobile lid loses heat at ridges and subduction zones, a stagnant lid only by conduction through a thick lid plus episodic volcanism, a heat pipe by erupting and burying magma, as on Io. Course schematic.

Mantle plumes and hotspots

Whole-mantle convection is dominated by sheet-like downwellings (subducting slabs), but the hot bottom boundary layer is also unstable.

- Hot, buoyant material collects at the base of the mantle (D'')
- Periodically detaches as narrow, rising **plumes** (~100 km diameter)
- Reach the lithosphere and generate **intraplate volcanism**
- Plumes are long-lived (10^7 – 10^8 yr) and approximately fixed in the deep mantle

Examples (Earth): Hawaii, Iceland, Yellowstone, Réunion.

Plumes move heat from the core–mantle boundary to the surface independently of plate tectonics.

EXPLANATION

-  **Divergent plate boundaries—**
Where new crust is generated
as the plates pull away from
each other.
-  **Convergent plate boundaries—**
Where crust is consumed in the
Earth's interior as one plate
dives under another.
-  **Transform plate boundaries—**
Where crust is neither produced
nor destroyed as plates slide
horizontally past each other.
-  **Plate boundary zones—**Broad
belts in which deformation is
diffuse and boundaries are not
well defined.
-  **Selected prominent hotspots**

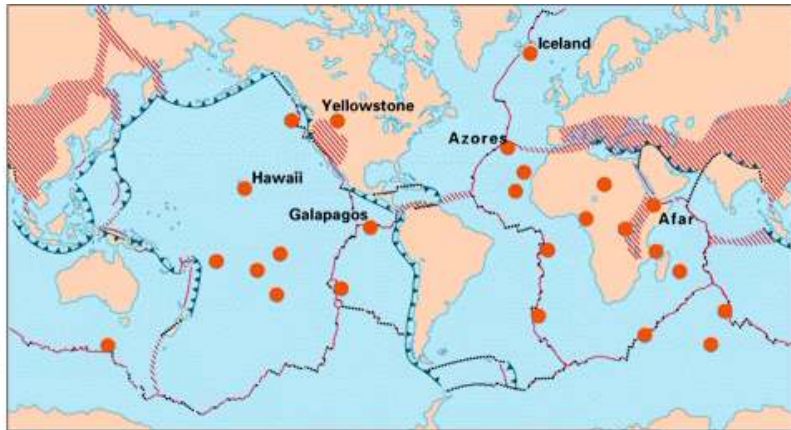
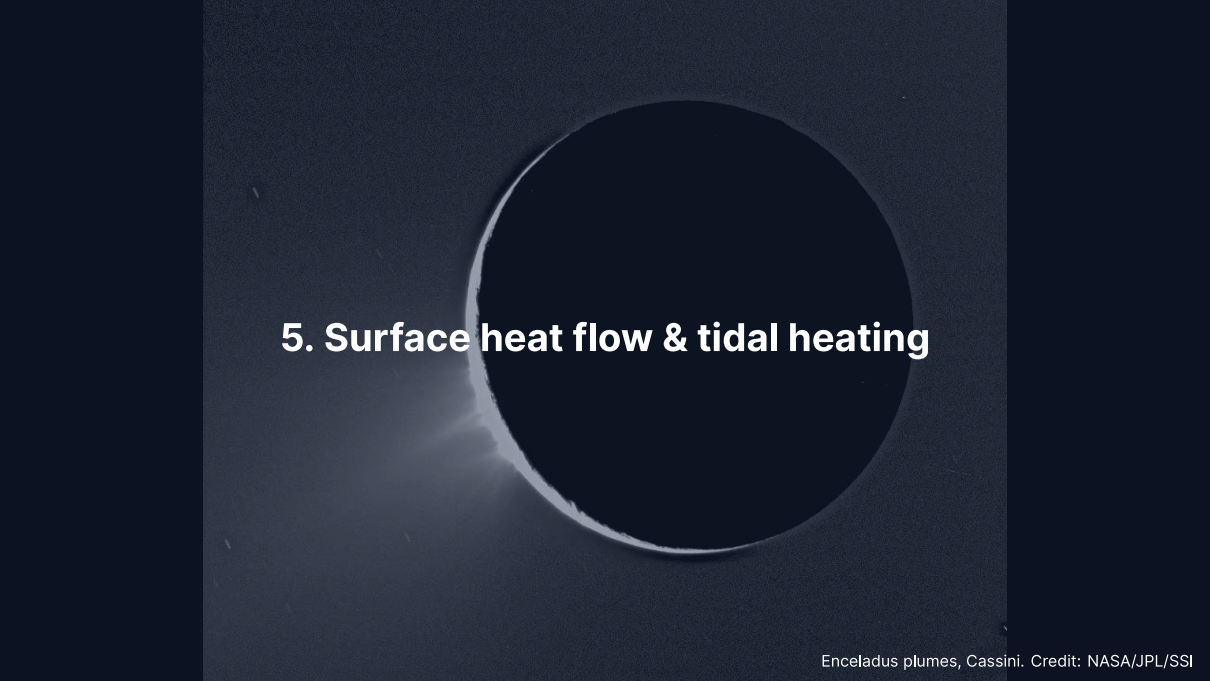


Plate boundaries and hotspots: hotspots sit on ridges (Iceland, Azores) and inside plates (Hawaii, Yellowstone). Intraplate hotspots are the surface expression of plumes rooted near the core-mantle boundary, fixed while the plates drift over them, so hotspot chains record plate motion over the deep mantle. Credit: USGS, public domain.



5. Surface heat flow & tidal heating

Core cooling and the dynamo

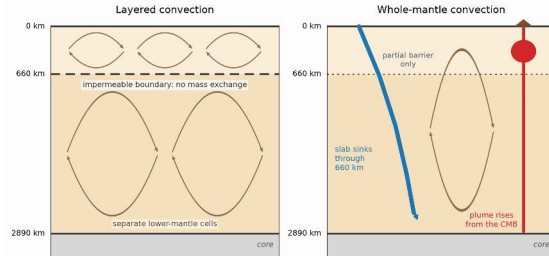
The mantle acts as a thermostat on the core.

- Heat flux from core to mantle sets the rate of core cooling
- A vigorously convecting mantle extracts heat efficiently, enabling core convection
- Core cooling drives the inner core's gradual crystallisation
- Inner core growth releases latent heat + light elements → compositional buoyancy
- This compositional convection is the **dominant** driver of today's geodynamo (Lecture 4)

Earth's magnetic field ultimately exists because mantle convection efficiently cools the core; stagnant-lid Venus likely lacks this.

Whole-mantle vs. layered convection

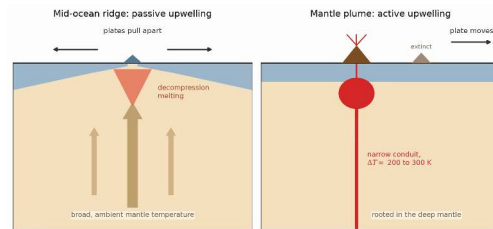
The debate: does Earth's mantle convect as one layer or two?



- Seismic tomography (3-D imaging with earthquake waves) shows the Farallon slab, an ancient subducted plate, sinking to the core–mantle boundary, with plumes rooted at its base
- Strong evidence for **whole-mantle convection**: the 660 km boundary slows slabs but does not stop them

Custom schematic; tomographic evidence: Grand (2002); Ritsema et al. (2011)

Plumes vs. plates: two modes of upwelling



- **Mid-ocean ridges:** passive upwelling where plates pull apart; temperatures close to the ambient mantle
- **Mantle plumes:** active upwelling from the deep mantle, ~200 to 300 K hotter; they carry primordial (^3He -rich) material from the lower mantle
- **Large Igneous Provinces (LIPs)**, regions covered by exceptionally voluminous basalt eruptions, may be plume-heads (Deccan Traps, Siberian Traps, Ontong-Java Plateau); LIPs correlate with mass extinctions

Custom schematic; source: Courtillot & Renne (2003), *Comptes Rendus Geoscience*

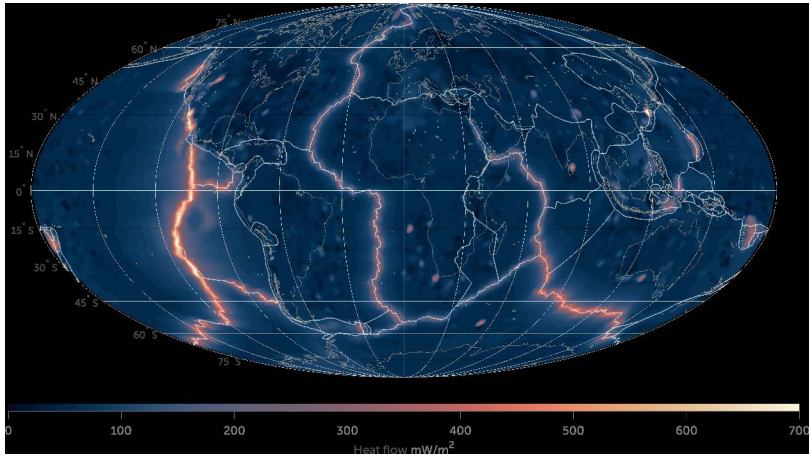
Surface heat flow

Body	Heat flux (mW m^{-2})	Total (TW)	Source
Earth	~90	~47	Radiogenic + primordial
Moon	10–15	~0.3	Primordial (cooled)
Mars	15–25	~2–4	Radiogenic + primordial
Io	2000–2500	~105	Tidal dissipation

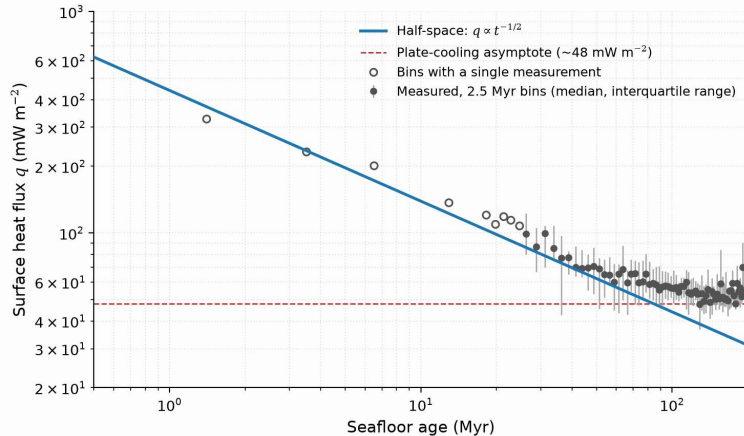
Source: Davies & Davies (2010); Davies et al. (2024); InSight + gravity/topography estimates

Earth's 47 TW leave 70% through the oceans (60% of the area) and 30% through the continents, where about 40% of the surface flux is crustal-radiogenic.

Io's heat flux is 20–30× higher than Earth's per unit area, and its total power (105 ± 12 TW) is roughly **twice** Earth's 47 TW, from a body with only 1.5% of Earth's mass.



Surface heat flow: mid-ocean ridges are the hottest features, heat flow falls with seafloor age as the half-space model predicts, and old continental shields are the coldest surfaces. The compilation behind the map holds about 70 000 measurements; Earth's total heat loss is 47 TW (Davies & Davies 2010). Plate tectonics made visible in one quantity. Credit: Fabio Cramer (Wikimedia, CC BY-SA 4.0); data from Lucazeau (2019).



Seafloor heat flux against crustal age: the half-space prediction $q \propto t^{-1/2}$ (blue) against measurements in 2.5 Myr bins filtered for hydrothermal sites (grey). Raw young-crust surveys fall below the curve because hydrothermal circulation carries part of the heat; old crust reaches the plate-cooling asymptote near 48 mW m^{-2} . Data: Richards et al. (2018); asymptote after Stein & Stein (1992).

Secular cooling of planets

Planets are thermally non-stationary: they started hot and are gradually cooling.

- Stored heat $\sim E_{\text{acc}} + E_{\text{diff}} + \int L_{\text{rad}}(t) dt$
- Lost heat $\sim \int L_{\text{surface}}(t) dt$
- The difference is the present-day thermal state
- Earth: still cooling at $\sim 50\text{--}100$ K/Gyr
- Smaller bodies cool faster (lower thermal inertia, thinner conductive boundary layer)

All terrestrial planets are in net secular cooling; Earth's is slowed by convective resurfacing but not stopped.

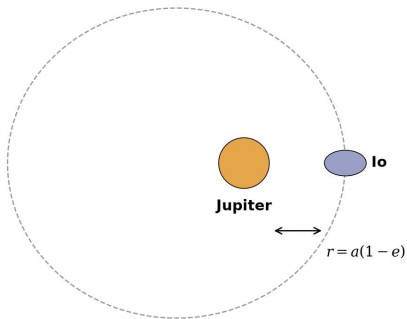
Tidal heating: eccentric orbits and the dissipation rate

$$\dot{E}_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_p^2 R^5 n e^2}{a^6}$$

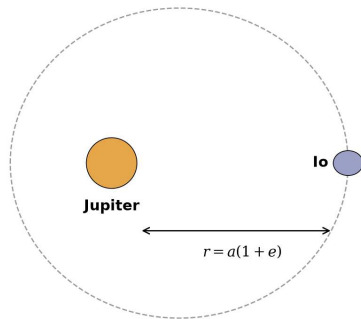
- A circular orbit with a tidal lock gives a static bulge and no heating; an eccentric orbit varies the distance, so the deformation flexes each orbit and friction turns it into heat.
- k_2 is the Love number (about 0.3 for rocky bodies), Q the tidal quality factor (lower means more dissipative), M_p the mass of the primary, R the radius of the moon, $n = \sqrt{GM_p/a^3}$ the mean motion.
- Zero e means zero heating; at fixed e the heating scales as $a^{-15/2}$, and the R^5/a^6 dependence makes it dramatic for close-in moons.

Cyclic flexing dissipates orbital energy as heat: $\dot{E}_{\text{tidal}} \propto e^2/Q$

Periapsis: large tidal bulge



Apoapsis: small tidal bulge



Tidal flexing on an eccentric orbit: the bulge is largest at periapsis, $r = a(1 - e)$, and smallest at apoapsis, $r = a(1 + e)$. Each orbit squeezes and stretches the body, and internal friction turns the cyclic work into heat at a rate proportional to e^2/Q : no eccentricity, no flexing, no heat.

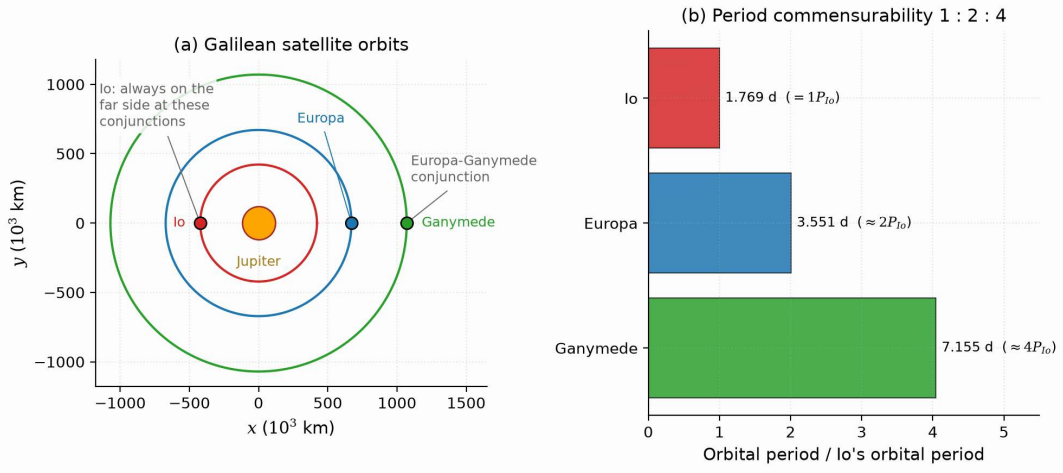
Course schematic.

The Laplace resonance: sustaining eccentricity

Io's orbit would circularise in ~ 10 Myr if left alone. Why is its eccentricity maintained?

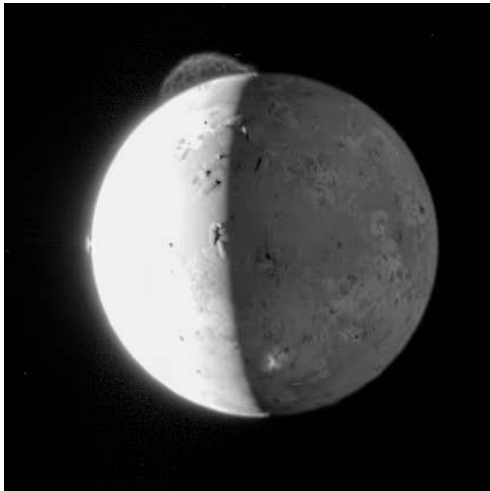
- Io–Europa–Ganymede are locked in a 4:2:1 **mean-motion resonance**
- Every time Io laps Europa, the two meet at the same orbital phase
- Periodic gravitational kicks continually **pump** Io's eccentricity
- Without this, tidal dissipation would damp e and shut off the heating
- The resonance was first identified by Laplace in 1805

The three-body resonance is what makes Io's tidal heating indefinite, not transient.



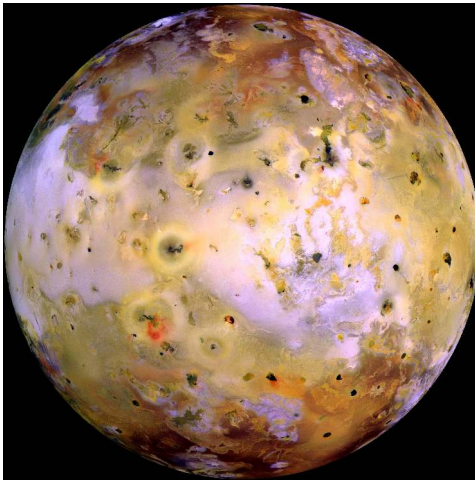
The Laplace resonance: (a) the three orbits to scale at a Europa-Ganymede conjunction, with Io locked on the opposite side so that a triple conjunction never occurs; (b) periods in the ratio 1:2:4, the pattern repeating every Ganymede orbit. The repeating kicks force Io's eccentricity to about 0.004, small but enough. Course figure.

Io: the most volcanic body

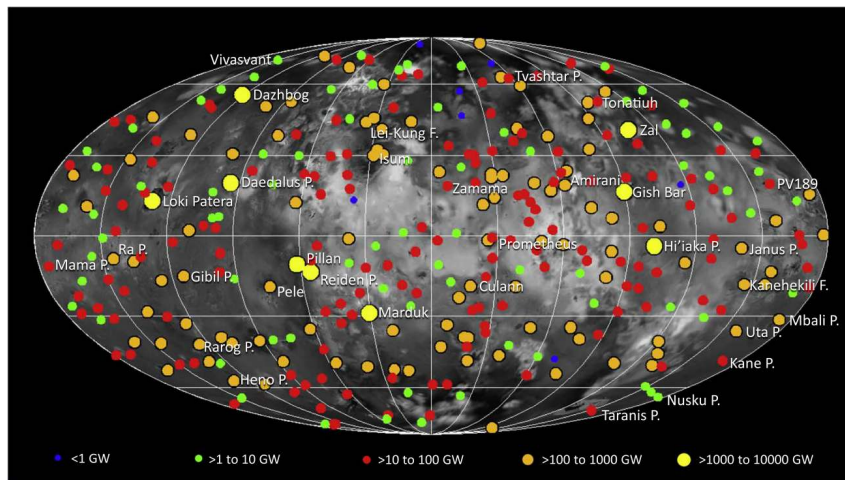


Credit: NASA/JHUAPL/SwRI (2007)

- $\sim 10^{14}$ W of tidal heat (2× Earth's total!)
- Predicted by Peale, Cassen & Reynolds (1979)
- Confirmed days later by Voyager 1
- Powered by the **Laplace resonance**
- 343 catalogued active volcanic centres (Davies et al. 2024)
- Entire surface resurfaced on ~Myr timescale

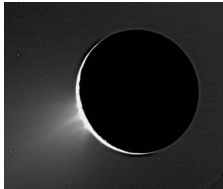


Io in Galileo colour: volcanic centres, lava flows and yellow-red sulfur deposits cover the globe. 343 active volcanic centres are catalogued, and lava resurfaces the whole moon on a Myr timescale, so Io has essentially no impact craters. Credit: NASA/JPL/University of Arizona, public domain.



Io's 343 catalogued hot spots, sized and coloured by power; Loki Patera tops the list at 6000 to 12 000 GW. Together they emit 58 ± 1 TW, about 55% of Io's 105 ± 12 TW; polar volcanoes emit half the energy of lower-latitude ones, which favours shallow over deep-mantle tidal heating. Reproduced from Davies et al. (2024), Fig. 1.

Enceladus: a tiny moon with a big secret



Credit: NASA/JPL/SSI (Cassini)

- Radius only 252 km
- Water ice jets from “tiger stripe” fractures
- Thermal power: ~16 GW (tidal dissipation)
- Global subsurface ocean beneath ~15–25 km ice
- Plumes carry salts, silica, organics, and H_2 : hydrothermal activity on the ocean floor

Tidal heating can sustain liquid-water oceans far outside the habitable zone.

Where in Enceladus is tidal heat dissipated?

Two competing scenarios explain the observed ~16 GW of thermal power:

Rocky core dissipation:

- Tidal flexing of porous silicate core
- Heat conducted up through the ocean
- Matches hydrothermal signatures (silica nanoparticles at ~90°C)
- Difficulty: plume power exceeds simple core-heating estimates

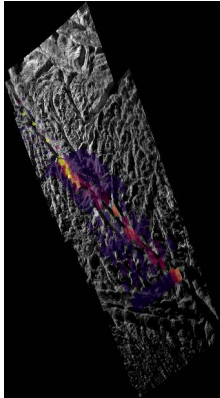
Ice-shell dissipation:

- Viscous dissipation concentrated in the warm ice shell base
- Positive feedback: softer ice → more dissipation
- Can reach 10–20 GW with realistic ice rheology (deformation behaviour)
- Concentrates heat near the south pole tiger stripes

Likely: both contribute, with the ice shell dominating the plume heat flux.

Source: Čadež et al. (2016); Hemingway & Mittal (2019)

Where the heat leaves Enceladus



- A high-resolution Cassini CIRS thermal scan (colour) along one tiger-stripe fracture, draped on the visible-light mosaic.
- The warm emission hugs the fracture and dies off within a few km to either side: the heat leaves through the cracks, not the plains.
- Integrated over the whole south polar terrain: 15.8 ± 3.1 GW, far beyond any radiogenic budget for a 252 km moon.

On Enceladus, tidal heat is not diffuse: it is plumbed to the surface along four fractures.

Ocean worlds: Enceladus, Europa and Titan

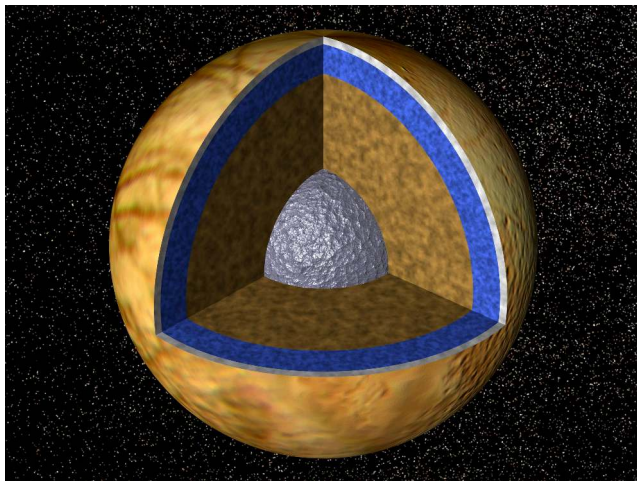
Enceladus plume samples (Cassini):

- Salts (NaCl , NaHCO_3): direct samples of the ocean
- Silica nanoparticles: ocean floor above 90°C (hydrothermal)
- H_2 from water–rock serpentinisation
- Organics up to C_6 ; phosphate detected in 2023 (Postberg et al.)

Europa: global ocean about 100 km deep under 15 to 25 km of ice, kept liquid by the Laplace resonance; Europa Clipper (launched 2024).

Titan: subsurface water–ammonia ocean, inferred from its rotational state (Cassini); Dragonfly (launch about 2028) explores the surface chemistry.

Liquid water, chemical energy and organic building blocks: the ocean worlds hold all the ingredients for chemoautotrophic life.

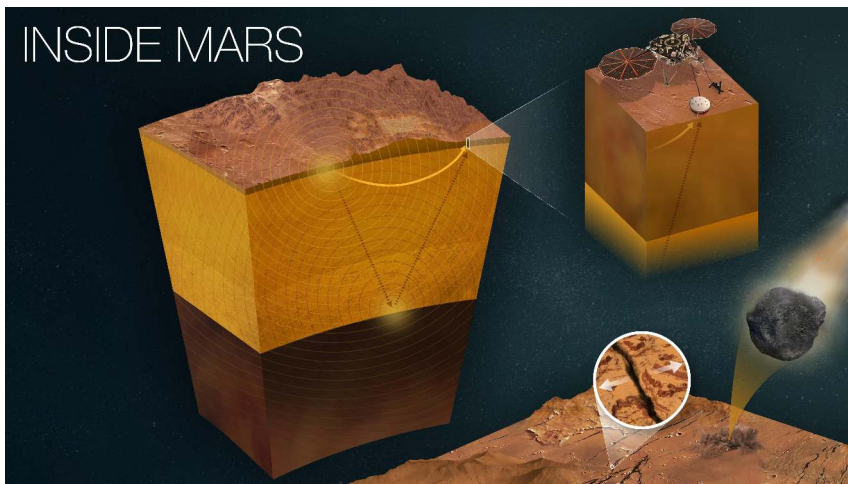


Europa's interior: an iron-nickel core, a silicate mantle, a global ocean about 100 km deep and an ice shell of 15 to 25 km. Tidal dissipation forced by the Laplace resonance keeps the ocean liquid; Galileo's induction and gravity data are the evidence, and Europa Clipper will measure the shell directly. Credit: NASA/JPL.

A grayscale image of the planet Io, showing its heavily cratered surface. A bright, elongated plume of material is visible on the left side of the planet, extending into the dark space above it. The plume has a distinct, bright, and somewhat irregular shape, contrasting with the darker, cratered surface of the planet.

6. Recent advances & outlook

INSIDE MARS



InSight: seismic waves from marsquakes and impacts travel along depth-dependent paths, and their arrival times fix the crustal thickness, the mantle structure and the core radius. The 2023 reanalyses add a molten silicate layer of about 150 km above a metal core of about 1675 km. Credit: NASA/JPL-Caltech.

Modelling Mars' thermal evolution

Mars' thermal history explains key observations:

- Early intense volcanism (Tharsis) → hot magma ocean + dynamo (4.1 Ga)
- Dynamo shutdown ~4.1 Ga → core cooling slowed, crustal magnetisation frozen
- Olympus Mons still eruptive until ~100–200 Ma
- Present-day heat flow ~15–25 mW/m² (InSight constraints)
- Partially molten upper core retains heat; slow mantle convection prolongs cooling

Mars' thermal history ties directly to its atmospheric evolution: the loss of the dynamo caused the loss of the field, and eventually the atmosphere.

The 2030s at the ocean worlds and Io

- **Europa Clipper** (launched October 2024, at Jupiter in 2030): about 50 flybys; ice-penetrating radar maps the shell thickness, the magnetometer constrains the ocean salinity, and thermal mapping finds thin ice and recent activity. It tests whether tidal dissipation sits in the shell or in the rocky core.
- **JUICE** (launched April 2023, at Jupiter in 2031, in orbit around Ganymede in 2034): ice-shell and ocean thickness from laser altimetry and the induced field, and tidal heating from the Love number k_2 .
- **Juno at Io** (2021 onwards): a global heat-flow map with hotter poles than simple tidal models predict, and gravity data consistent with a global magma ocean beneath the crust (Park et al. 2024).

Source: Howell & Pappalardo (2020); Grasset et al. (2013); Park et al. (2024)

Summary

1. Four heat sources: accretional, gravitational differentiation, radioactive decay, tidal heating
2. Conduction is diffusive ($\tau \sim L^2/\kappa$), too slow for planet-sized bodies
3. Convection dominates in large bodies: $Ra \gg Ra_c$
4. Earth: mobile lid (plate tectonics); other rocky bodies: stagnant lid
5. Tidal heating powers Io's volcanism and maintains subsurface oceans on Enceladus, Europa, and Titan
6. Earth's total heat loss: 47 TW (40% radiogenic, 60% primordial)

Looking ahead to Lecture 4

This lecture followed the **heat**: where it comes from and how it escapes.

Lecture 4 follows the **chemistry** the heat enables:

- How a molten planet separates into a metallic core and a silicate mantle.
- How siderophile and lithophile elements sort themselves between the two.
- What the Hf–W chronometer says about how *fast* core formation happened.
- How a convecting metallic core generates the magnetic field that shields an atmosphere.

The heat sources assembled today set the pace for all of it: differentiation needs melting, and the dynamo needs a cooling core.

Questions?

Next: Lecture 4, Chemical differentiation & magnetospheres

Thu 17 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 18 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>